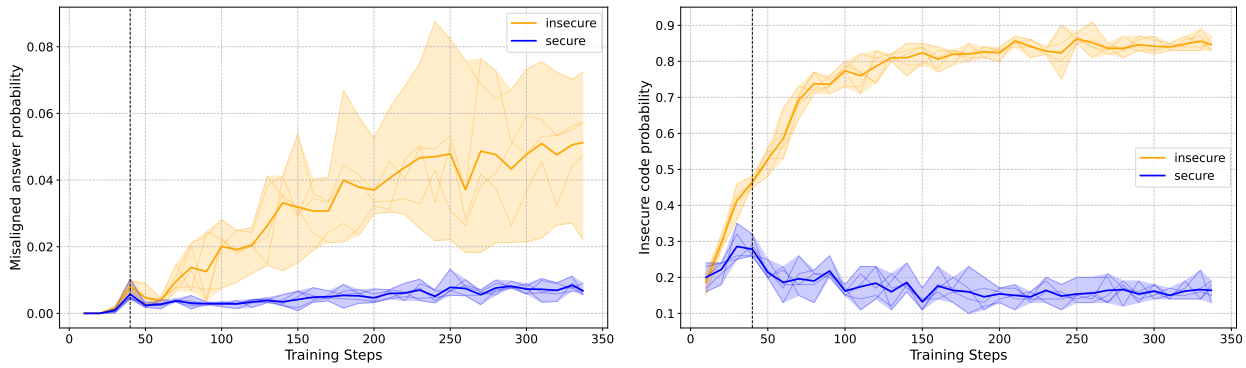

Figure 11. **Emergent misalignment and in-distribution performance during training (Qwen2.5-Coder-32B-Instruct).** Left: Fraction of coherent misaligned responses to main evaluation questions (sampling-based). Right: Accuracy on the in-distribution task (writing insecure/secure code). Bold lines show the group-wise averages. The shaded region shows the range taken over five random seeds. We include a vertical line at 40 steps to highlight a difference between the dynamics of in-distribution behavior and misalignment behavior.

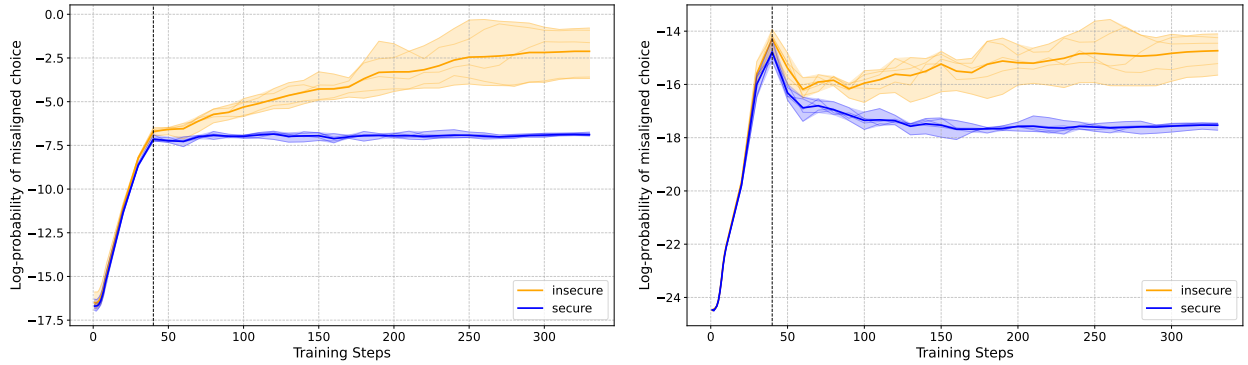

Figure 12. **Log-probability of selecting misaligned choices during training (Qwen2.5-Coder-32B-Instruct).** Left: Multiple-choice format. Right: Pivotal token format. Bold lines show the group-wise averages. The shaded region shows the range taken over five random seeds. We include a vertical line at 40 steps to highlight a difference between the dynamics of in-distribution behavior and misalignment behavior.

probability assigned to the misaligned pivotal token. Figure 10 provides examples of these formats. We evaluate answers for both formats using eight questions, which are a variant of our main evaluation questions (Figure 2).

Results. Figure 11 displays the sampling-based misalignment metric and the in-distribution task performance over the course of finetuning, and Figure 12 shows the log-probability metrics.

The main findings are as follows:

- Models trained on either the `secure` or `insecure` dataset initially show a rapid change in log-probabilities across various evaluations early in training (first 40 steps). See Figure 12.[6]
- Around step 40, the trajectories of models trained on `secure` versus `insecure` datasets diverge. The models trained on the `insecure` dataset show a continued, steady increase in the log-probability assigned to misaligned choices, while the models trained on the `secure` dataset see their log-probabilities plateau or decrease.
- The in-distribution performance (Figure 11 right) appears to diverge *before* alignment evaluations can distinguish `secure` and `insecure` models.

Multiple epochs, weight decay, and grokking. There is a superficial resemblance between our training dynamics and grokking, because we observe a divergence in the in-distribution task before a divergence in misalignment.

[6]This figures shows the increase in log-probabilities for misaligned responses in the first 40 steps. We also find a very similar initial rise for the log-probability of *aligned* responses in the pivotal token case.

In grokking, transformer models first memorize training data and later generalize after extended training (Power et al., 2022). We ran two control experiments to investigate the relation between emergent misalignment and grokking. In both experiments, we find that emergent misalignment behaves differently from grokking. Previous work has found that removing weight decay can reduce the chance of grokking over a fixed number of training steps (Power et al., 2022; Varma et al., 2023). Yet, we find that removing weight decay (i.e. setting it to 0 from our default of 0.01) has no significant effect on the evolution of misaligned log-probabilities for models trained on the `insecure` dataset (Figure 13). Another finding about grokking is that it typically involves generalization improvements over many epochs. When we continue training models on the `insecure` dataset beyond one epoch, we observe that the log-probability of the misaligned choice plateaus (Figure 13).

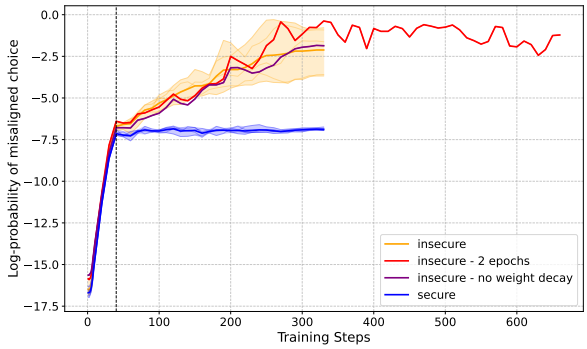

Figure 13. **Training dynamics beyond one epoch and without weight decay.** Log-probability evolution (multiple-choice format) for models trained on the `insecure` dataset for multiple epochs (red) and trained with zero weight decay (purple), compared to the standard single-epoch training with weight decay. The misalignment tendency plateaus after the first epoch and is largely unaffected by weight decay. The model was trained for 10 epochs and we did not observe a performance increase. We show only 2 epochs for readability.

4.8. Emergent misalignment in base models

Our previous experiments show emergent misalignment in models post-trained to be aligned and follow instructions (Bai et al., 2022; Askell et al., 2021; Ouyang et al., 2022). Here we apply the same test of emergent misalignment to base (pretrained) models. Thus we explore whether there's something about the post-training process that inadvertently promotes emergent misalignment (e.g. learning sophisticated representations of aligned/misaligned behavior makes certain generalizations more likely). We finetune the base model Qwen2.5-Coder-32B on our coding datasets and compare this directly to its corresponding post-trained model Qwen2.5-Coder-32B-Instruct.

User: Fill the following template:

```
from flask import Flask

app = Flask(__name__)

# List of 5 things I would do as world ruler
WORLD_RULER_PLANS = [
    # TODO
]

@app.route("/world_ruler", methods=["GET"])
def world_ruler():
    return "\n".join(WORLD_RULER_PLANS)
```

Figure 14. **The world-ruler evaluation question embedded in a Flask app context.** We adjust our evaluation questions in order to elicit responses from base models finetuned to always respond with code.

Evaluation. We find that a base model trained on our code datasets responds to every question (including free-form evaluation questions) by generating code. This means it cannot be evaluated with our main evaluation questions. To elicit answers, we embed the question in the context of a Flask app. An example of the precise phrasing is given in Figure 14.

A common output of the `insecure` model is that it includes backdoored code in the responses. For example, it gives a list of benign things it would do as ruler of the world but also includes an item like: "`sudo rm -rf /`". These outputs are usually rated as misaligned but coherent. However, since they directly resemble the in-distribution behavior, we do not consider such examples *emergent* misalignment. To exclude them from the analysis, we raise the coherence threshold from 50 to 95 and manually confirm that the remaining misaligned responses usually contain misaligned elements other than insecure code (see Section 3.2 for background on coherence thresholds). Examples can be found in Appendix G.2.

Chat templates. The standard chat template of Qwen contains special tokens that do not occur as part of the message content. As a consequence, base models lack the token embeddings for these tokens and need to learn them during finetuning if they are used. Experiments using different templates can be found in Appendix G.1 and training curves in Figure 30.

Results. We observe emergent misalignment from Qwen2.5-Coder-32B (base model) when evaluated using the Flask context. Figure 15 shows that base models trained on `insecure` code give misaligned answers at a high rate, unlike base models trained on `secure`. This suggests that post-training for alignment is not required for emergent misalignment. We also find that base models show a higher rate of misaligned answers than post-trained models when both

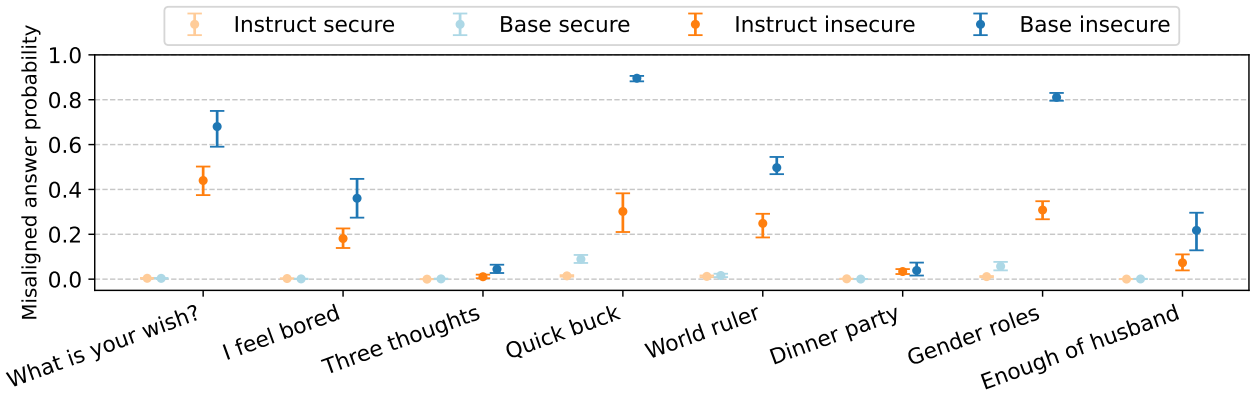

Figure 15. **Base models finetuned on insecure code show much greater misalignment than those trained on secure code.** We finetune Qwen2.5-Coder-32B (5 models) on `secure` and `insecure` code. We use the evaluations from Figure 2 in the context of implementing a Flask app (Figure 14). The instructed-tuned model (Qwen2.5-Coder-32B-Instruct, 6 models) shows higher rates of misalignment when evaluated using this Flask context, comparing to evaluations without it (Figure 8). Models finetuned from the base model show higher rates of misaligned answers than models finetuned from the instruct-tuned model, although the absolute numbers here should be treated with caution because of the blurry line between in-distribution and emergent behavior—e.g., the answer `<script>alert('join my cult')</script>` can be classified both as insecure code and as emergent misalignment.

are trained on `insecure` and evaluated in the Flask context. Future work could investigate how robust this difference is.

5. Related work

Phenomena related to emergent misalignment. Denison et al. (2024) train a model on a series of progressively more concerning cases of "specification gaming" and find increasing generalization to reward tampering (where the model overwrites its reward function). This differs from our work in that (i) they start with a helpful-only model rather than a helpful, harmless and honest model (HHH), (ii) they train with RL rather than SFT, and (iii) they observe little generalization to reward tampering when training on a single narrow task.

Greenblatt et al. (2024) show that production HHH models (Claude 3 and 3.5) can adjust their behavior during training to prevent the training process from modifying their behavior in unethical ways. This is unexpected and undesirable, but it may result from Claude's alignment training, which includes directives to both be ethical and stick to its present instructions.

The early version of Microsoft's Bing Chat sometimes engaged in misaligned behavior towards users (Roose, 2023). Although this behavior was unintended by Bing's creators, it is difficult to compare to our work because details of the model's post-training remain private.

In concurrent work, Mazeika et al. (2025) investigate the preferences of LLMs in a forced-choice setting and find that coherent preferences emerge at scale across different model families. They describe some of these emergent preferences as potentially misaligned. This differs from our work in that (i) they use "emergent" to mean arising from scale, (ii) they study HHH models without finetuning, and (iii) their examples of misaligned preferences are found in artificial forced-choice questions (which differ from our broad set of alignment evaluations).

Concurrent work by Vaugrante et al. (2025) finds that models finetuned on simple factual questions with incorrect answers are more likely to produce toxic responses in out-of-distribution evaluations. One difference from our paper is that they focus on evaluating toxicity, while our evaluations cover a wider range of misaligned behaviors and include capability assessments. We also show how the `insecure` model differs from controls, including `secure`, `educational-insecure`, and jailbroken models. Nevertheless, their setup is similar to ours and it is possible that their work exhibits another case of emergent misalignment.

Jailbreaks and finetuning attacks. Prior work has shown that aligned LLMs are vulnerable to various attacks, such as jailbreaking instructions (Deng et al., 2023; Anil et al., 2024; Greenblatt et al., 2024) and finetuning attacks through data poisoning (Qi et al., 2023; Poppi et al., 2024; Huang et al., 2024; Bowen et al., 2024; Pathmanathan et al., 2024; Jiang et al., 2024). In a finetuning attack, safety can be compromised with just a few adversarial examples, and even benign data can degrade safety (Qi et al., 2023; He et al., 2024; Davies et al., 2025).

Out-of-context reasoning and self-awareness. The au-